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OSPF Quiz 2

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Result of OSPF Quiz 2:

Total Questions	Full Score	Passing Rate	Your Score	Correct Answer Percentage	Elapsed
8	100	80%	100	100%	00:00:54

Congratulations!

You passed this test!

If you want to retake this quiz, please press Ctrl + F5 on Windows or press CMD + R on Mac.

Your answers are shown below:

Question 1

Refer to the exhibit. Which configuration issue is preventing the OSPF neighbor relationship from being established between the two routers?



R1#show running-config Building configuration… ! interface GigabitEthernet1/0 mtu 1600 ip address 192.168.0.1 255.255.255.252 negotiation auto ! router ospf 1 router-id 1.1.1.1 passive-interface default no passive-interface GigabitEthernet1/0 network 192.168.0.1 0.0.0.0 area 0 !	R2#show running-config Building configuration… ! interface GigabitEthernet2/0 ip address 192.168.0.2 255.255.255.252 negotiation auto ! router ospf 1 router-id 2.2.2.2 passive-interface default no passive-interface GigabitEthernet2/0 network 192.168.0.2 0.0.0.0 area 0 !

- A. R1 has an incorrect network command for interface Gi1/0
- B. R2 is using the passive-interface default command
- C. R2 should have its network command in area 1

- **D. R1 interface Gi1/0 has a larger MTU size**

Question 2

Refer to the exhibit. What is the next hop address for traffic that is destined to host 10.0.1.5?

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

Gateway of last resort is not set

C       1.0.0.0/8 is directly connected, Loopback0
        10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O          10.0.1.3/32 [110/100] via 10.0.1.3, 00:40:07, Serial0
C          10.0.1.0/24 is directly connected, Serial0
O          10.0.1.5/32 [110/5] via 10.0.1.50, 00:40:07, Serial0
O          10.0.1.4/32 [110/10] via 10.0.1.4, 00:40:07, Serial0
```

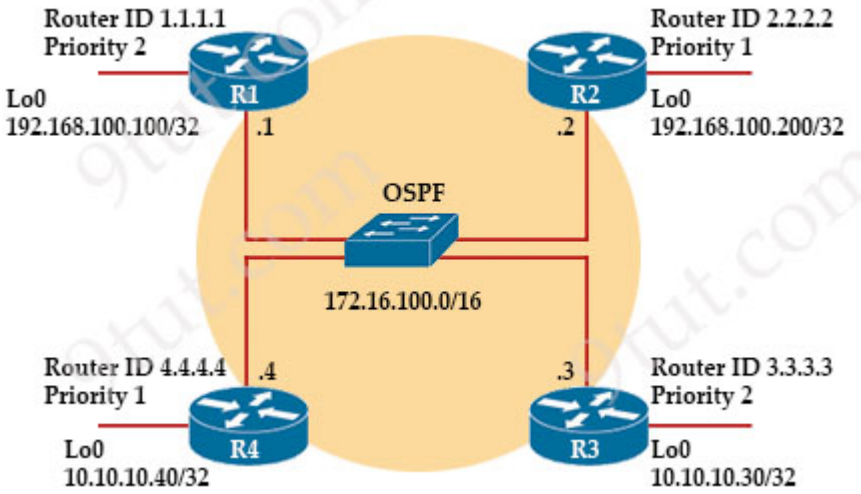
- **A. 10.0.1.50**
- B. 10.0.1.3
- C. Loopback 0
- D. 10.0.1.4

Explanation

Traffic destined to host 10.0.1.5 will match the entry “O 10.0.1.5/32 …” so the next-hop address will be 10.0.1.50.

Question 3

Refer to the exhibit. If the switch reboots and all routers have to re-establish OSPF adjacencies, which routers will become the new DR and BDR?



- **A. Router R3 will become the DR and router R1 will become the BDR.**
- B. Router R4 will become the DR and router R3 will become the BDR.
- C. Router R1 will become the DR and router R2 will become the BDR.
- D. Router R3 will become the DR and router R2 will become the BDR.

Explanation

After the new election, R3 and R1 have highest priority (of 2) so they will be elected DR and BDR. R3 will be elected DR because its has higher router ID (of 3.3.3.3).

Question 4

Which step in the link-state routing process is described by a router sending Hello packets out all of the OSPF-enabled interfaces?

- A. electing the designated router
- **B. establishing neighbor adjacencies**
- C. injecting the default route
- D. exchanging link-state advertisements

Question 5

Which two minimum parameters must be configured on an active interface to enable OSPFv2 to operate? (Choose two)

- A. OSPF stub flag
- **B. OSPF process ID**
- **C. OSPF area**
- D. OSPF MD5 authentication key
- E. IPv6 address

Explanation

To configure OSPFv2 under interface mode we must use the command “ip ospf {process-id} area {area-id}”. For example:
R1(config-if)#ip ospf 1 area 2

Question 6

Refer to the exhibit. Refer to the exhibit. After the configuration is applied, the two routers fail to establish an OSPF neighbor relationship. what is the reason for the problem?



Router1(config)#interface GigabitEthernet1/1 Router1(config-if)#description ***Connection to Router2*** Router1(config-if)#ip address 10.10.10.1 255.255.255.252 Router1(config-if)#ip ospf hello-interval 5 Router1(config)#router ospf 1000 Router1(config-router)#router-id 1.1.1.1 Router1(config-router)#network 10.10.10.0 0.0.0.3 area 0	Router2(config)#interface GigabitEthernet1/1 Router2(config-if)#description ***Connection to Router1*** Router2(config-if)#ip address 10.10.10.2 255.255.255.252 Router2(config)#router ospf 1001 Router2(config-router)#router-id 2.2.2.2 Router2(config-router)#network 10.10.10.0 0.0.0.3 area 0 Router2(config-router)#passive-interface default Router2(config-router)#no passive-interface GigabitEthernet1/1

- A. The OSPF router IDs are mismatched
- **B. Router2 is using the default hello timer**
- C. The network statement on Router1 is misconfigured
- D. The OSPF process IDs are mismatched

Question 7

An engineer must configure an OSPF neighbor relationship between router R1 and R3. The authentication configuration has been configured and the connecting interfaces are in the same 192.168.1.0/30 subnet. What are the next two steps to complete the configuration? (Choose two)

- A. configure the hello and dead timers to match on both sides
- **B. configure the interfaces as OSPF active on both sides**
- C. configure the same router ID on both routing processes
- **D. configure both interfaces with the same area ID**
- E. configure the same process ID for the router OSPF process

Explanation

We don’t need to configure hello and dead timers for OSPF neighbor as OSPF can auto negotiate the network type (with the same hello and dead timers on both sides). We just have to make sure both interfaces are running OSPF and they are in the same area ID.

Question 8

Refer to the exhibit. What action establishes the OSPF neighbor relationship without forming an adjacency?

```
R1#show ip ospf int gi0/0
Gig0/0/0 is up, line protocol is up
  Internet Address 10.201.24.8/28, Area 1, Attached via Network Statement
  Process ID 100, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID   Cost   Disabled   Shutdown   Topology Name
          0         1         no         no         Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 10.201.24.8
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:05

R2#show ip ospf int gi0/0
Gig0/0/0 is up, line protocol is up
  Internet Address 10.201.24.1/28, Area 1
  Process ID 100, Router ID 172.16.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 172.16.1.1, Interface address 10.201.24.1
  No backup designated router on this network
  Timer intervals configured, Hello 20, Dead 80, Wait 40, Retransmit 5
```

- A. modify priority
- **B. modify hello interval**
- C. modify process ID
- D. modify network type

Explanation

In this question, R1 & R2 will not establish OSPF neighbor relationship because the hello & dead intervals are not the same.
If we modify the hello & dead intervals to the same values then R1 & R2 will become OSPF neighbors. But they will establish OSPF adjacency too.

Answer 'modify process ID' is not correct as the process ID is only locally significant.

Answer 'modify hello interval' is partially correct as we need to change both “hello and dead” interval, not “hello” interval only.

Answer 'modify priority' is a good answer because if we change both the priorities to 0 then they cannot elect DR/BDR so the adjacency cannot be complete.

Answer 'modify network type' is a good answer too because if we change one of the network type to “point to point” then OSPF neighbor relationship can still come up but adjacency cannot (no routes are getting exchanged).

But two good answers are only correct after fixing the hello & dead intervals to the same value (so that OSPF neighbor relationship can come up) so answer 'modify hello interval' should be the best choice here.

Note:

Neighbors
Routers that share a common segment become neighbors on that segment. Neighbors are elected via the Hello protocol. Hello packets are sent periodically out of each interface using IP multicast (Appendix B). Routers become neighbors as soon as they see themselves listed in the neighbor’s Hello packet.

Two routers will become neighbors if they agree on the following: **Same Area ID, same authentication (if used), same hello & dead intervals, same subnet mask and same stub area flag**.

Adjacencies
Adjacency is the next step after the neighboring process. Adjacent routers are routers that go beyond the simple Hello exchange and proceed into the database exchange process.

The adjacency building process takes effect after multiple stages have been fulfilled. Routers that become adjacent will have the exact link-state database. The states an interface passes through before becoming adjacent to another router are: **Down -> Attempt (optional) -> Init -> 2-Way -> Exstart -> Exchange -> Loading -> Full**.

At Full state the adjacency is complete.

Reference: https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/7039-1.html#t22

-> Two OSPF neighbors do not exchange any routing information – the only packets they exchange is Hello packets.

Adjacencies on Point-to-Point Interfaces
OSPF will always form an adjacency with the neighbor on the other side of a point-to-point interface such as point-to-point serial lines. There is no concept of DR or BDR. The state of the serial interfaces is point to point.

Comments (10) Comments

1. hans
   January 2nd, 2022

   Q.3 the best answer should be C and E

   Q4 . the answer should be both C and D

2. R.F.
   January 14th, 2022

   Just a side-note:

   Did you realize that the Loopback-Addresses are partially wrong in the following question?
   It reads two times 1092.x.x.x instead of 192.x.x.x

   ‘Refer to the exhibit. If the switch reboots and all routers have to re-establish OSPF adjacencies, which routers will become the new DR and BDR?’

3. 9tut
   January 15th, 2022

   @R.F.: Thanks for your detection! We updated the exhibit!

4. Mubassir
   June 24th, 2023

   @9Tut
   Q7- Refer to the exhibit. What is the next hop address for traffic that is destined to host 10.0.1.5?

   Answer options does not include the correct answer, please correct it.

   The options to choose are:
   A. 10.0.1.4
   B. 10.0.1.50

C. 10.0.1.3
D. Loopback 0

Where as the correct answer is expected to be 10.0.1.5, which is not enlisted in the answer options whereas, can be seen in routing table. Also, the option 10.0.1.50 is not in the routing table.

Can you please update @9Tut ?

5. coodyscoops
   February 1st, 2024

   the answers for Q3 should be C and E.

   The reason being is that the keyword is neighbor relationship. I would assume that OSPF would be active already if he is trying to configure OSPF neighborship on the interfaces but if the hello amd dead timers do not match the neighborship would never form, and if we cant assume that OSPFis active on the interfaces he wants to become neighbors, then that question has 3 correct answers.

6. Refer to the exhibit. What is the next hop address for traffic that is destined to host 10.0.1.5?
   April 15th, 2024

   why is it 10.0.1.50? they are asking about HOST 10.0.1.5 and O 10.0.1.5/32 is a network learned by OSPF and the 1st address available on that network (10.0.1.5/32) is 10.0.1.6. so it should use O 10.0.1.4/32 network where the 1st HOST address on that network is 10.0.1.5.

7. 9tut
   April 16th, 2024

   10.0.1.5 is the destination address but the next-hop IP address for this network is 10.0.1.50.

8. CCNP in progress
   June 20th, 2024

   An engineer must configure an OSPF neighbor relationship between router R1 and R3. The authentication configuration has been configured and the connecting interfaces are in the same 192.168.1.0/30 subnet. What are the next two steps to complete the configuration? (Choose two)

   By default interfaces are active so why should I need to configure it?

9. Thomas
   September 11th, 2024

   Are these questions accurate to the ones that will be asked in the real exam? Are there any that are pulled directly from the current exam set of questions?

10. abdou
    May 17th, 2025

    je ne comprends pas pourquoi ce n’est pas le D pour la question 8.

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- NAT DHCP SSH Sim
- ==== Lab Sims for CCNA (although they are rather old but please learn all of them too) ====
- IP Services Sim Version 2
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- LACP Configuration Sim
- Voice VLAN Configuration Sim
- VLAN and Trunking Configuration Sim
- Named Access-list & Port Security Sim
- Named Access-list & DHCP Snooping Sim
- VLAN and CDP Sim
- IPv4 and IPv6 Deployment Sim
- Static Routing Configuration Sim 3
- VLAN & LLDP Sim

- IPv6 Static Routing Sim
- Named Access-list & DHCP Snooping Sim 2
- ISL Trunking and LACP Sim
- === Below Sims are old and retired so you do not need to learn them ===
- Static Routing Configuration Sim 2
- IP Services Sim
- Static Routing Configuration Sim

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